

MACKALL AAF, NC, USA

WMO: 723034

Lat: 35.033N

Lon: 79.500W

Elev: 376

StdP: 14.50

Time Zone: -5.00 (NAE)

Period: 07-19

WBAN: 93747

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 18.0	(c) 21.9	(d) 0.7	(e) 5.8	(f) 27.9	(g) 5.7	(h) 7.5	(i) 28.9	(j) 22.8	(k) 63.1	(l) 19.5	(m) 54.2	(n) 3.3	(o) 340	(p) 0.413

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.6	95.3	75.3	93.4	74.9	91.2	74.4	79.5	88.3	78.0	87.4	77.0	86.3	8.4	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.7	145.9	82.2	75.5	135.3	81.1	74.6	131.1	80.4	43.1	88.4	41.6	87.4	40.5	86.8	89.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.6	16.4	14.1	10.0	98.7	6.5	2.8	5.4	100.7	1.6	102.4	-2.1	104.0	-6.8	106.0
			9.1	79.9	6.4	1.0	4.5	80.6	0.7	81.2	-2.9	81.7	-7.5	82.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	61.8	40.6	45.9	51.5	62.3	70.6	78.1	80.8	78.9	73.8	62.5	49.7	46.1	(6)	(7)
	DBStd	15.97	10.57	10.12	9.70	7.87	6.23	4.55	3.44	3.80	5.37	8.16	8.21	9.81	(7)	(8)
	HDD50	930	327	185	102	6	0	0	0	0	0	6	106	198	(8)	(9)
	HDD65	3095	758	539	429	143	26	0	0	0	5	145	461	589	(9)	(10)
	CDD50	5248	35	71	150	374	639	844	956	897	714	393	97	78	(10)	(11)
	CDD65	1937	1	4	12	61	200	394	491	432	269	67	2	4	(11)	(12)
	CDH74	18369	3	33	128	620	1786	4019	4902	3927	2313	613	14	11	(12)	(13)
	CDH80	7773	0	1	12	150	623	1883	2305	1701	936	162	0	0	(13)	(14)
(14) Wind	WSAvg	5.8	5.7	6.6	6.9	7.4	6.5	5.8	5.5	4.7	5.2	5.3	5.2	5.1	(14)	(15)
(15) Precipitation	PrecAvg	46.8	3.7	3.3	4.1	3.2	3.1	4.6	5.1	5.0	5.0	3.3	3.1	3.2	(15)	(16)
	PrecMax	62.3	6.7	6.4	7.3	6.1	7.2	12.5	8.8	10.0	14.8	9.1	8.4	6.8	(16)	(17)
	PrecMin	33.0	0.9	1.6	1.2	1.0	1.3	0.8	1.9	1.3	1.3	0.0	0.7	1.5	(17)	(18)
	PrecStd	6.5	1.8	1.4	1.6	1.3	1.4	2.8	2.2	1.9	3.2	2.4	2.0	1.3	(18)	(19)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	73.2	78.2	82.0	87.4	93.8	99.1	98.0	95.5	94.0	88.9	76.5	75.5	(19)	(20)
		MCWB	61.6	64.1	62.2	66.5	72.2	75.3	76.1	75.4	71.4	73.6	63.9	67.5	(20)	(21)
	2%	DB	67.0	73.2	78.1	84.3	89.3	95.2	95.3	93.3	91.1	84.0	72.6	70.9	(21)	(22)
		MCWB	59.7	61.1	61.9	65.7	69.7	74.7	76.5	75.5	72.5	70.7	62.9	65.2	(22)	(23)
	5%	DB	62.9	68.3	73.9	81.0	86.6	92.6	93.3	91.0	88.8	80.5	69.2	67.3	(23)	(24)
		MCWB	56.3	58.4	59.2	64.8	69.2	73.8	76.3	75.3	72.9	69.1	61.0	63.4	(24)	(25)
	10%	DB	58.4	63.8	69.1	77.5	83.4	89.7	90.9	88.8	86.1	76.8	65.7	63.2	(25)	(26)
		MCWB	52.7	56.1	58.0	62.8	68.7	73.2	75.7	74.7	72.5	67.8	58.4	58.9	(26)	(27)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	65.7	66.6	66.5	72.0	75.0	78.4	83.5	79.3	85.8	78.4	69.1	69.7	(27)	(28)
		MCDB	69.8	73.8	74.7	79.5	84.2	90.9	89.3	89.8	87.0	80.5	71.6	71.9	(28)	(29)
	2%	WB	62.1	64.0	64.9	69.8	73.1	77.1	80.0	78.2	78.0	75.0	66.3	66.7	(29)	(30)
		MCDB	65.9	69.8	73.1	77.3	81.6	89.2	89.0	87.8	82.7	79.2	69.7	69.6	(30)	(31)
	5%	WB	57.9	61.3	62.9	67.8	71.9	75.9	78.3	77.3	75.5	72.9	63.4	63.9	(31)	(32)
		MCDB	60.9	66.8	69.5	75.5	81.3	87.4	88.1	86.2	83.0	78.8	67.0	66.4	(32)	(33)
	10%	WB	53.0	57.0	60.8	65.8	70.8	74.9	77.3	76.4	74.0	70.0	60.0	59.7	(33)	(34)
		MCDB	57.1	63.1	67.2	74.2	79.7	85.5	87.0	84.8	81.1	74.4	64.6	62.6	(34)	(35)
(35) Mean Daily Temperature Range	5% DB	MDBR	21.5	22.5	23.0	24.2	21.7	21.3	19.6	18.8	20.9	23.2	24.5	21.2	(35)	(36)
		MCDBR	26.7	27.2	28.2	27.2	25.4	24.1	22.2	21.5	24.9	25.8	27.7	22.5	(36)	(37)
	5% WB	MCWBR	19.2	16.9	13.9	11.8	9.1	7.5	7.0	6.7	9.5	13.3	18.1	16.7	(37)	(38)
		MCWBR	22.3	24.0	22.7	20.4	20.0	21.7	19.8	18.8	18.4	20.7	20.4	19.6	(38)	(39)
(39) Clear-Sky Solar Irradiance	taub		0.315	0.327	0.362	0.401	0.459	0.490	0.538	0.513	0.427	0.389	0.335	0.312	(39)	(40)
		taud	2.502	2.462	2.396	2.283	2.165	2.120	2.005	2.064	2.300	2.373	2.484	2.545	(40)	(41)
	Ebn at Noon		283	292	288	281	265	255	242	246	264	265	272	277	(41)	(42)
		Edn at Noon	26	30	36	42	48	50	56	51	39	33	27	24	(42)	(43)
(43) All-Sky Solar Radiation	RadAvg		843	1050	1392	1744	1917	2023	1915	1739	1476	1208	924	731	(43)	(44)
	RadStd		50	121	104	122	154	142	120	82	127	154	75	62	(44)	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (71 neighbors)	+0.86	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+292	+211

Nomenclature: See separate page